PEICD / C++ REVIEW BOOK

C++ 期中
程式練習複習本

原題圖片 × 中文題目 × 我的程式碼

24	27	6
實際完成的題目	完整原題圖片	涵蓋章節（Ch1–Ch6）

題目來源：Absolute C++, 6th Edition（原題圖片）
中文翻譯來源：PEICD 期中練習題單
程式碼來源：PEICD 練習紀錄（每題最新提交版本）
所有入選題目之最新評測紀錄：AC • 100 分

REVIEW EDITION / 2026.11.04

複習目錄

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> 每題依序包含原書題目截圖、完整繁體中文翻譯與你最後提交的程式碼。保留原程式的運算式、命名、註解與內部空白行；僅略去程式碼開頭及結尾的空白行，並將換行格式統一以利閱讀。

CHAPTER 01 / PROGRAMMING PROJECT 06 · 01 / 24

計程車費率

Ch1 • Programming Project 6 AC 100 / 100

01 原題圖片 • Original Problem

6. A city cab service provider charges customers based on distance and time. The service charges $2 for the first two kilometers. After the first two kilometers, the service charges $0.50 per kilometer for the next six kilometers. After six kilometers, the customer is charged $1 per kilometer. At the end of the ride, a fee of $0.2 per minute is charged based on the total travel duration. Write a program that reads the distance traveled in kilometers and the time taken in minutes for a cab ride and computes the fare based on the service's charges.

課本原題 • 第 73 頁

02 中文題目翻譯

一家市區計程車業者依據距離與時間向乘客收費。前兩公里收費　$2。前兩公里之後，接下來六公里每公里收費 $0.50。超過六公里後，每公里收費 $1。行程結束時，再依總行車時間按每分鐘 $0.2 收費。撰寫程式，讀入一次搭乘的距離（公里）與所花時間（分鐘），並依該業者的收費方式計算車資。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp — C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;
int main() {
    double k, t, re = 0;
    cin >> k >> t;
    if (k <= 2) {
        re = 2;
    } else if (k <= 8) {
        re = 2 + (k - 2) * 0.50;
    } else {
        re = 2 + 6 * 0.50 + (k - 8) * 1;
    }
    re += t * 0.2;

    cout << fixed << setprecision(2) << re;
}
```

CHAPTER 01 / PROGRAMMING PROJECT 08 · 02 / 24

巴比倫平方根近似

Ch1 · Programming Project 8 AC 100 / 100

01 原題圖片 · Original Problem

8. The Babylonian algorithm to compute the square root of a positive number **n** is as follows:
   1. Make a **guess** at the answer (you can pick n/2 as your initial guess).
   2. Compute ***r*** = **n / guess**.
   3. Set **guess** = **(guess + r) / 2**.
   4. Go back to step 2 for as many iterations as necessary. The more steps 2 and 3 are repeated, the closer **guess** will become to the square root of **n**.

課本原題 · 第 73 頁 · 第 1/2 張

Write a program that inputs a double for **n**, iterates through the Babylonian algorithm five times, and outputs the answer as a `double` to two decimal places. Your answer will be most accurate for small values of **n**.

課本原題 · 第 74 頁 · 第 2/2 張

02 中文題目翻譯

使用 Babylonian algorithm（巴比倫演算法）計算正數 n 的平方根，其方法如下：

1. 先猜一個答案 guess（初始猜測值可以選 n/2）。
2. 計算 r = n / guess。
3. 令 guess = (guess + r) / 2。
4. 回到步驟 2，依需要重複迭代。步驟 2 與 3 重複越多次，guess 就會越接近 n 的平方根。

撰寫程式，輸入 double 型別的 n，將巴比倫演算法迭代五次，再將 double 型別的答案輸出至小數點後兩位。當 n 較小時，答案會最精確。

03 我的 C++ 程式碼

Latest submitted source · syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;
int main() {
    double n;
    cin >> n;
    double g = n / 2;
    for (int i = 0; i < 5; i++) {
        double r = n / g;
        g = (g + r) / 2;
    }
    cout << fixed<<setprecision(2)<<g;
}
```

CHAPTER 01 / PROGRAMMING PROJECT 13 · 03 / 24

咖啡因單位換算

Ch1 • Programming Project 13 AC 100 / 100

01 原題圖片 • Original Problem

13. Scientists estimate that consuming roughly 10 grams of caffeine at once is a lethal overdose. Write a program that inputs the number of milligrams of caffeine in a drink and outputs how many of those drinks it would take to kill a person. A 12-ounce can of cola has approximately 34 mg of caffeine, while a 16-ounce cup of coffee has approximately 160 mg of caffeine.

課本原題 • 第 74 頁

02 中文題目翻譯

科學家估計，一次攝取約 10 公克咖啡因是致命的過量劑量。撰寫程式，輸入一杯飲料含有的咖啡因毫克數，並輸出需要多少杯這種飲料才會使一人死亡。一罐 12 盎司的可樂約含 34 毫克咖啡因；一杯 16 盎司的咖啡約含 160 毫克咖啡因。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;
int main() {
    double n;
    cin >> n;
    cout << fixed << setprecision(8) << 10000 / n;
}
```

CHAPTER 02 / PROGRAMMING PROJECT 02 · 04 / 24

每月提款與利息

Ch2 • Programming Project 2 AC 100 / 100

01 原題圖片 • Original Problem

2. In order to manage their funds efficiently, a person deposits an amount of money in a bank account. They withdraw a fixed portion on the fifth day of every month for their household expenses. The bank credits the interest accumulated on their balance on the last day of every month. Write a program that inputs the initial deposit made, the interest rate provided by bank, and the fixed amount withdrawn every month. Calculate the number of months the amount (including the credited interest) can be utilized. (*Hint*: Use a loop)

   Note: Check for boundary conditions – If the interest earned every month is greater than the fixed withdrawal amount, the program should print that message and return.

課本原題 • 第 125 頁

02 中文題目翻譯

為了有效管理資金，某人在銀行帳戶中存入一筆錢，並於每個月的第五天提領固定金額作為家庭開銷。銀行在每個月最後一天，將帳戶餘額累積的利息入帳。撰寫程式，輸入初始存款、銀行提供的利率，以及每月提領的固定金額。計算這筆錢（包括入帳的利息）可以使用多少個月。（提示：使用迴圈。）

注意：檢查邊界情況。如果每個月賺取的利息大於固定提款金額，程式應印出這項訊息並返回。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;
int main() {
    double a, b, c, oa;
    cin >> a >> b >> c;
    b = b / 100;
    oa = a;
    int t = 0;
    while (1) {
        if (a <= 0 || a<c) {
            cout << t;
            break;
        }
        a -= c;
        t += 1;
        a += a * b;
        if (oa <= a) {
            cout << "INFINITE";
            break;
        }
    }
}
```

CHAPTER 02 / PROGRAMMING PROJECT 04 · 05 / 24

3 到 100 的質數

Ch2 • Programming Project 4 AC 100 / 100

01 原題圖片 • Original Problem

4. Write a program that finds and prints all of the prime numbers between 3 and 100. A prime number is a number that can only be divided by one and itself (i.e., 3, 5, 7, 11, 13, 17, . . .).

   One way to solve this problem is to use a doubly-nested loop. The outer loop can iterate from 3 to 100, while the inner loop checks to see whether the counter value for the outer loop is prime. One way to decide whether the number *n* is prime is to loop from 2 to *n* – 1; if any of these numbers evenly divides *n*, then *n* cannot be prime. If none of the values from 2 to *n* – 1 evenly divide *n*, then *n* must be prime. (Note that there are several easy ways to make this algorithm more efficient.)

課本原題 • 第 125 頁

02 中文題目翻譯

撰寫程式，找出並印出 3 到 100 之間的所有質數。質數是只能被 1 和自己整除的數，例如 3、5、7、11、13、17、……。

一種解法是使用雙層巢狀迴圈。外層迴圈從 3 走到 100，內層迴圈檢查外層計數器的值是否為質數。檢查數字 n 是否為質數的一種方法，是從 2 迭代到 n − 1；如果其中任何數能整除 n，則 n 不可能是質數。如果 2 到 n − 1 之間沒有任何數能整除 n，則 n 必為質數。（注意：有數種方法可以讓這個演算法更有效率。）

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int isp(int x){
    for(int i=2;i<x;i=i+1){
        if(x%i==0)return 0;
    }
    return 1;
}

int main() {
    for(int i=3;i<=100;i++){
        if(isp(i)){
            cout<<i<<"\n";
        }
    }
}
```

CHAPTER 02 / PROGRAMMING PROJECT 07 · 06 / 24

平均成績點數

Ch2 • Programming Project 7 AC 100 / 100

01 原題圖片 • Original Problem

7. Write a program to calculate the grade point average (GPA) of a student by getting their grades for all their subjects. Grades and their grade points are

```
Grade S          10 points
Grade A           9 points
Grade B           8 points
Grade C           7 points
Grade D           6 points
Grade E           5 points
Grade F           0 points
```

The program should input the number of subjects, followed by the grade obtained in each subject. The program should calculate the average of the grade points and print it.

課本原題 • 第 126 頁

02 中文題目翻譯

撰寫程式，取得學生所有科目的成績，並計算其平均成績點數（GPA）。各成績及對應點數如下：

成績與點數

成績	點數
S	10
A	9
B	8
C	7
D	6
E	5
F	0

程式應先輸入科目數，再輸入各科得到的成績。計算各成績點數的平均值並印出。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    map<string,double>mp;
```

1.cpp C++17 / VS Code Dark+

```cpp
    int n;
    cin>>n;
    mp["S"]=10;
    mp["A"]=9;
    mp["B"]=8;
    mp["C"]=7;
    mp["D"]=6;
    mp["E"]=5;
    mp["F"]=0;
    double sum =0;
    for(int i=0;i<n;i++){
        string s;
        cin>>s;
        sum+=mp[s];
    }
    cout<<fixed<<setprecision(8)<<sum/n;
}
```

CHAPTER 02 / PROGRAMMING PROJECT 08 · 07 / 24

攝氏與華氏相同的溫度

Ch2 • Programming Project 8 AC 100 / 100

01 原題圖片 • Original Problem

8. Write a program that finds the temperature, as an integer, that is the same in both Celsius and Fahrenheit. The formula to convert from Celsius to Fahrenheit is as follows:

$$\text{Fahrenheit} = \frac{9}{5}\text{Celsius} + 32$$

Your program should create two integer variables for the temperature in Celsius and Fahrenheit. Initialize the temperature to 100 degrees Celsius. In a loop, decrement the Celsius value and compute the corresponding temperature in Fahrenheit until the two values are the same.

課本原題 • 第 126 頁

02 中文題目翻譯

撰寫程式，找出攝氏與華氏數值相同的整數溫度。攝氏轉換為華氏的公式如下：

> Fahrenheit = (9/5) Celsius + 32

程式應建立兩個整數變數，分別表示攝氏與華氏溫度。最初將攝氏溫度設為 100 度。使用迴圈遞減攝氏值，並計算對應的華氏溫度，直到兩個溫度值相同。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    double c = 100;
    for (int i = c;; i--) {
        // if( 5 * (i-32) == 9 * i){
        if (i == (int)((9 / 5.0) * i + 32)) {
            cout << i;
            break;
        }
    }
}
```

CHAPTER 03 / PROGRAMMING PROJECT 01 · 08 / 24

影片檔案大小

Ch3 · Programming Project 1 AC 100 / 100

01 原題圖片 · Original Problem

1. A megabyte is 8000 kilobits. Write a program that will read in the bit rate in kilobits per second and the duration in minutes of a video clip and will output the file size of the video clip in megabytes.

   Define a function to compute the file size of the video clip. Your program should use a globally defined constant for the number of kilobits per megabyte.

課本原題 · 第 169 頁

02 中文題目翻譯

1 megabyte 等於 8000 kilobits。撰寫程式，讀入影片的位元率（kilobits per second）與片長（分鐘），並輸出影片檔案大小（megabytes）。

定義一個函式來計算影片的檔案大小。程式應使用全域定義的常數，表示每 megabyte 有多少 kilobits。

03 我的 C++ 程式碼

Latest submitted source · syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    double a, b;
    cin >> a >> b;

    cout <<fixed<<setprecision(8)<< a * b*60 / 8000.0;
}
```

CHAPTER 03 / PROGRAMMING PROJECT 05 · 09 / 24

服裝尺寸計算

Ch3 • Programming Project 5 AC 100 / 100

01 原題圖片 • Original Problem

5. Write a program that asks for the user's height, weight, and age, and then computes clothing sizes according to the following formulas.
   - Hat size = weight in pounds divided by height in inches and all that multiplied by 2.9.
   - Jacket size (chest in inches) = height times weight divided by 288 and then adjusted by adding one-eighth of an inch for each 10 years over age 30. (Note that the adjustment only takes place after a full 10 years. So, there is no adjustment for ages 30 through 39, but one-eighth of an inch is added for age 40.)
   - Waist in inches = weight divided by 5.7 and then adjusted by adding one-tenth of an inch for each 2 years over age 28. (Note that the adjustment only takes place after a full 2 years. So, there is no adjustment for age 29, but one-tenth of an inch is added for age 30.)

   Use functions for each calculation. Your program should allow the user to repeat this calculation as often as he or she wishes.

課本原題 • 第 170 頁

02 中文題目翻譯

撰寫程式，詢問使用者的身高、體重與年齡，再依下列公式計算服裝尺寸：

- 帽子尺寸 = 體重（磅）除以身高（英吋），再乘以 2.9。
- 夾克尺寸（胸圍，英吋）= 身高乘以體重，再除以 288；超過 30 歲後，每滿 10 年加上 1/8 英吋。（注意：必須滿 10 年才調整。因此 30～39 歲不調整；40 歲時增加 1/8 英吋。）
- 腰圍（英吋）= 體重除以 5.7；超過 28 歲後，每滿 2 年加上 1/10 英吋。（注意：必須滿 2 年才調整。因此 29 歲不調整；30 歲時增加 1/10 英吋。）

每項計算都使用函式。程式應允許使用者依意願重複計算。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    double a, b, cc;
    double x, y, z;
    while (cin >> a >> b >> cc) {
        x = b / a * 2.9;

        double c = cc;
        y = a * b / 288;
        if (c > 30) {
```

1.cpp C++17 / VS Code Dark+

```
            c -= 30;
            y += (int)(c / 10) * (1 / 8.0);
        }

        c = cc;
        z = b / 5.7;
        if (c > 28) {
            c -= 28;
            z += (int)(c / 2) * 1 / 10.0;
        }

        cout << fixed << setprecision(8) << x << " " << y << " " << z << "\n";
    }
}
```

CHAPTER 03 / PROGRAMMING PROJECT 10 · 10 / 24

估計孩子身高

Ch3 • Programming Project 10 AC 100 / 100

01 原題圖片 • Original Problem

10. One way to estimate the height of a child is to use the following formula, which uses the height of the parents:

$$H_{\text{male_child}} = ((H_{\text{mother}}\ 13/12) + H_{\text{father}})/2$$

$$H_{\text{female_child}} = ((H_{\text{father}}\ 12/13) + H_{\text{mother}})/2$$

All heights are in inches. Write a function that takes as input parameters the gender of the child, height of the mother in inches, and height of the father in inches, and outputs the estimated height of the child in inches. Embed your function in a program that allows you to test the function over and over again until telling the program to exit. The user should be able to input the heights in feet and inches, and the program should output the estimated height of the child in feet and inches. Use the integer data type to store the heights.

課本原題 • 第 171 頁

02 中文題目翻譯

估計孩子身高的一種方法，是使用以下依據父母身高的公式：

> H_male_child = ((H_mother 13/12) + H_father) / 2 H_female_child = ((H_father 12/13) + H_mother) / 2

所有身高皆以英吋為單位。撰寫函式，以孩子的性別、母親身高（英吋）與父親身高（英吋）作為輸入參數，輸出估計的孩子身高（英吋）。將此函式放入一個程式中，讓使用者可以反覆測試函式，直到要求離開程式。使用者應能以英呎與英吋輸入身高，程式也應以英呎與英吋輸出估計的孩子身高。使用整數資料型別儲存身高。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp — C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    string s;
    double a,b,c,d;
    while(cin>>s>>a>>b>>c>>d){

        double H_mother,H_father;
        H_mother = a*12+b;
        H_father = c*12+d;
        double H_male_child = ((H_mother* 13.0/12) + H_father) / 2.0;
        double H_female_child = ((H_father* 12.0/13) + H_mother) / 2.0;
        int re = (s=="F")?H_female_child:H_male_child;
        cout<<re/12<<" "<<re%12<<"\n";
    }

```

1.cpp C++17 / VS Code Dark+

```
18 }
```

CHAPTER 03 / PROGRAMMING PROJECT 13 · 11 / 24

抽出四位得獎者

Ch3 • Programming Project 13 AC 100 / 100

01 原題圖片 • Original Problem

13. You have four identical prizes to give away and a pool of 25 finalists. The finalists are assigned numbers from 1 to 25. Write a program to randomly select the numbers of 4 finalists to receive a prize. Make sure not to pick the same number twice. For example, picking finalists 3, 15, 22, and 14 would be valid but picking 3, 3, 31, and 17 would be invalid, because finalist number 3 is listed twice and 31 is not a valid finalist number.

課本原題 • 第 174 頁

02 中文題目翻譯

你有四個相同的獎品要送出，以及 25 位決選者。決選者編號為 1～25。撰寫程式，隨機選出 4 位決選者的編號得獎，注意不要把同一個編號選到兩次。例如，選出 3、15、22、14 有效；選出 3、3、31、17 則無效，因為編號 3 被列出兩次，而且 31 不是有效的決選者編號。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    srand(time(0));
    set<int> s;
    int w = 0;
    while (w < 4) {
        int f = rand() % 25 + 1;
        if (!s.count(f)) {
            w++;
            cout << f << "\n";
            s.insert(f);
        }
    }
}
```

CHAPTER 04 / PROGRAMMING PROJECT 03 · 12 / 24

麵包店禮券

Ch4 • Programming Project 3 AC 100 / 100

01 原題圖片 • Original Problem

3. You have a voucher for a bakery, which you redeem to buy baked goods of your choice. The value of the voucher is fixed and the total bill of the items selected should not exceed the voucher value. If the voucher has any value left after the purchase, then it cannot be utilized again. Write a program that displays 4 items and their prices as a menu and then reads the value of the voucher.

   Write a function, with parameters `itemCode` and `voucherValue` passed by reference, that deducts the price of an item from the voucher's value only if the item's value is less that or equal to the voucher value. Use a loop to invoke the above function as many times as required until the value of the voucher is less than or equal to the cheapest item on the menu.

   Finally, display the calculated sum of the prices of the items selected, and the unutilized balance on the voucher after purchase.

課本原題 • 第 210 頁

02 中文題目翻譯

你有一張麵包店禮券，可以用來兌換喜歡的烘焙商品。禮券面額固定，所選商品總額不得超過禮券面額。如果購買後還有餘額，該餘額不能再次使用。撰寫程式，顯示 4 種商品及其價格作為選單，並讀入禮券面額。

撰寫函式，以參照方式傳入 itemCode 與 voucherValue 參數。只有當商品價格小於或等於禮券餘額時，才從禮券面額中扣除該商品價格。使用迴圈視需要呼叫上述函式，直到禮券餘額小於或等於選單中最便宜商品的價格。

最後，顯示所選商品價格的總和，以及購買後禮券未使用的餘額。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

vector<int> v = {0, 2, 3, 4, 5};

void f(double &itemCode ,double &voucherValue ){
    if ( 1<=itemCode && itemCode<=4  && voucherValue >= v[itemCode]) {
        voucherValue -= v[itemCode];
    }
}

int main() {
    double n;
    cin >> n;
    double on = n;
    double it;

    while (cin >> it) {
```

1.cpp C++17 / VS Code Dark+

```
        if(n<=2)break;
        f(it,n);
    }
    cout << R"(1 Muffin 2
2 Bagel 3
3 Croissant 4
4 Cake 5
)";
    cout << fixed << setprecision(2) << on-n << " " <<n ;
}
```

CHAPTER 04 / PROGRAMMING PROJECT 07 · 13 / 24

磅與盎司轉公制

Ch4 • Programming Project 7 AC 100 / 100

01 原題圖片 • Original Problem

7. Write a program that will read a weight in pounds and ounces and will output the equivalent weight in kilograms and grams. Use at least three functions: one for input, one or more for calculating, and one for output. Include a loop that lets the user repeat this computation for new input values until the user says he or she wants to end the program. There are 2.2046 pounds in a kilogram, 1000 grams in a kilogram, and 16 ounces in a pound.

課本原題 • 第 211 頁

02 中文題目翻譯

撰寫程式，讀入以磅與盎司表示的重量，並輸出等值的公斤與公克。至少使用三個函式：一個負責輸入、一個負責計算、一個負責輸出。加入迴圈，讓使用者可以用新輸入值重複計算，直到使用者表示想結束程式。1　公斤等於 2.2046 磅，1 公斤等於 1000 公克，1 磅等於 16 盎司。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp　　C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    double a, b;
    while (cin >> a >> b) {
        double bo = a + b / 16;
        double g = bo / 2.2046 * 1000;
        double q = g;
        while (q >= 1000) {
            q -= 1000;
        }
        cout << fixed << setprecision(8) << (int)(g / 1000) << " " << q << "\n";
    }
}

/*

1 kg = 2.2046 b
1000 g = 2.2046 b

2.2046/1000 b = g


*/
```

CHAPTER 04 / PROGRAMMING PROJECT 08 · 14 / 24

公制轉磅與盎司

Ch4 • Programming Project 8 AC 100 / 100

01 原題圖片 • Original Problem

8. Write a program like that of the previous exercise that converts from kilograms and grams into pounds and ounces. Use functions for the subtasks.

課本原題 • 第 211 頁

02 中文題目翻譯

撰寫與前一題類似的程式，將公斤與公克轉換為磅與盎司。各子工作使用函式處理。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    double a, b;
    while (cin >> a >> b) {
        double kg = a + b/1000;
        double bo = kg * 2.2046;

        double u = (bo-(int)(bo)) * 16;

        cout << fixed << setprecision(6) << (int)(bo) << " " << u << "\n";
    }
}

/*

1 kg = 2.2046 b
1000 g = 2.2046 b

2.2046/1000 b = g


*/
```

CHAPTER 04 / PROGRAMMING PROJECT 14 · 15 / 24

時光機的分鐘差

Ch4 • Programming Project 14 AC 100 / 100

01 原題圖片 • Original Problem

14. Your time machine is capable of going forward in time up to 24 hours. The machine is configured to jump ahead in minutes. To enter the proper number of minutes into your machine, you would like a program that can take a start time and an end time and calculate the difference in minutes between them. The end time will

課本原題 • 第 213 頁 • 第 1/2 張

always be within 24 hours of the start time. Use military notation for both the start and end times (e.g., 0000 for midnight and 2359 for one minute before midnight).

Write a function that takes as input a start time and an end time represented as an `int`, using military notation. The function should return the difference in minutes as an integer. Write a driver program that calls your subroutine with times entered by the user.

Hint: Be careful of time intervals that start before midnight and end the following day.

課本原題 • 第 214 頁 • 第 2/2 張

02 中文題目翻譯

你的時光機可以往未來移動，最長 24 小時，並設定為以分鐘數跳躍。為了輸入正確的分鐘數，你想要一個程式，接收起始時間與結束時間，並計算兩者相差幾分鐘。結束時間一定在起始時間之後的 24 小時內。起訖時間都使用軍用時間表示，例如午夜為 0000，午夜前一分鐘為 2359。

撰寫函式，接收以軍用時間表示的整數起始時間與結束時間，並傳回整數分鐘差。撰寫驅動程式，以使用者輸入的時間呼叫這個函式。

提示：注意起始於午夜前、結束於隔天的時間區間。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    int a, b;
    cin >> a >> b;

    int ta = a / 100 * 60 + a % 100;
    int tb = b / 100 * 60 + b % 100;

    if (ta > tb) {
        tb += 24 * 60;
    }

    cout << tb - ta << "\n";
}
```

1.cpp C++17 / VS Code Dark+

```
17
18 /*
19
20 1 kg = 2.2046 b
21 1000 g = 2.2046 b
22
23 2.2046/1000 b = g
24
25
26 */
```

CHAPTER 05 / PROGRAMMING PROJECT 04 · 16 / 24

母音出現次數

Ch5 • Programming Project 4 AC 100 / 100

01 原題圖片 • Original Problem

4. Write a program that reads in an array of characters. You may assume that there are fewer than 50 entries in the array. Your program determines how many entries are used. The output is to be a two-column list. The first column is a list of vowels in the English alphabet; the second column is the count of the number of occurrences of each vowel. The list should be sorted on entries in the first column in alphabetical order.

For the array values

```
b r e a d , b u t t e r, a n d j a m o n y o u r p l a t e
```

the output should be

```
Vowel      Count
  a          4
  e          3
  i          0
  o          2
  u          2
```

課本原題 • 第 266 頁

02 中文題目翻譯

撰寫程式，讀入一個字元陣列。可以假設陣列少於 50 個元素。程式決定使用了多少種不同的項目。輸出為兩欄清單：第一欄列出英文字母中的母音，第二欄列出各母音的出現次數。第一欄應依字母順序排列。

對於以下陣列值：

```
b r e a d , b u t t e r , a n d j a m o n y o u r p l a t e
```

輸出應為：

```
Vowel  Count
  a      4
  e      3
  i      0
  o      2
  u      2
```

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
```

1.cpp C++17 / VS Code Dark+

```cpp
    string s;
    getline(cin,s);

    vector<int> v(200);
    for(auto i:s){
        v[i] ++;
    }

    cout<<"Vowel Count\n";
    cout<<"a "<<v['a']<<"\n";
    cout<<"e "<<v['e']<<"\n";
    cout<<"i "<<v['i']<<"\n";
    cout<<"o "<<v['o']<<"\n";
    cout<<"u "<<v['u']<<"\n";


}
```

CHAPTER 05 / PROGRAMMING PROJECT 08 · 17 / 24

生日悖論模擬

Ch5 • Programming Project 8 AC 100 / 100

01 原題圖片 • Original Problem

8. The birthday paradox is that there is a surprisingly high probability that two people in the same room happen to share the same birthday. By birthday, we mean the same day of the year (ignoring leap years), but not the exact birthday including the birth year or time of day. Write a program that approximates the probability that two people in the same room have the same birthday, for 2 to 50 people in the room.

   The program should use simulation to approximate the answer. Over many trials (say, 5000), randomly assign birthdays to everyone in the room. Count up the number of times at least two people have the same birthday, and then divide by the number of trials to get an estimated probability that two people share the same birthday for a given room size.

   Your output should look something like the following. It will not be exactly the same due to the random numbers:

   For 2 people, the probability of two birthdays is about 0.002

   For 3 people, the probability of two birthdays is about 0.0082

   For 4 people, the probability of two birthdays is about 0.0163

   . . .

   For 49 people, the probability of two birthdays is about 0.9654

   For 50 people, the probability of two birthdays is about 0.969

課本原題 • 第 267 頁

02 中文題目翻譯

生日悖論指出，同一個房間內兩人生日相同的機率高得令人驚訝。這裡的生日是指一年中的同一天（忽略閏年），不是包含出生年份的完整日期。撰寫程式，對房間人數 2～50 人，近似估計其中有兩人生日相同的機率。

程式應使用模擬來近似答案。進行許多次試驗，例如 5000 次，每次隨機指定房間內每個人的生日。計算至少兩人生日相同的試驗次數，再除以試驗總次數，得到該房間人數下，兩人生日相同機率的估計值。

輸出應類似下列內容。由於亂數的緣故，不會完全相同：

```
For 2 people, the probability of two birthdays is about 0.002
For 3 people, the probability of two birthdays is about 0.0082
For 4 people, the probability of two birthdays is about 0.0163
. . .
For 49 people, the probability of two birthdays is about 0.9654
For 50 people, the probability of two birthdays is about 0.969
```

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
```

1.cpp C++17 / VS Code Dark+

```cpp
using namespace std;

int main() {
    srand(time(0));

    for (int i = 2; i <= 50; i++) {
        double sum = 0;
        for (int t = 0; t < 5000; t++) {
            vector<int> v(400);
            double same = 0;
            for (int k = 0; k < i; k++) {
                v[rand() % 365 + 1]++;
            }

            for (int it : v) {
                if (it >= 2) {
                    same = 1;
                }
            }

            sum += same;
        }
        cout << fixed << setprecision(6) << i << " " << sum / 5000 << "\n";
    }
}
```

CHAPTER 05 / PROGRAMMING PROJECT 10 · 18 / 24

冰淇淋庫存與購物車

Ch5 • Programming Project 10 AC 100 / 100

01 原題圖片 • Original Problem

10. Write a program that displays the list of flavors available at an ice-cream parlor with a unique code for each flavor, as shown below.

```
Code        Flavor
1           Chocolate
2           Vanilla
3           Strawberry
4           Raspberry
5           Butterscotch
```

The program then displays the code and quantity available for customers to purchase as shown below.

```
1           20
2           15
3           15
4           20
5           20
```

The program should read the user's choice of flavors and quantity and display the products remaining in the stock, until the stock is exhausted or the user signals that the program should end. Finally, the user's cart should be displayed as shown below

```
Chocolate      3
Strawberry     1
Raspberry      1
```

課本原題 • 第 268 頁

02 中文題目翻譯

撰寫程式，顯示冰淇淋店販售的口味清單，每種口味有唯一代碼，如下所示：

```
Code    Flavor
1       Chocolate
2       Vanilla
3       Strawberry
4       Raspberry
5       Butterscotch
```

接著程式顯示顧客可購買的代碼及數量，如下所示：

```
1      20
2      15
3      15
4      20
5      20
```

程式應讀取使用者選擇的口味與數量，並顯示庫存中剩餘的商品，直到庫存用盡，或使用者表示程式應結束。最後顯示使用者的購物車，如下所示：

```
Chocolate   3
Strawberry  1
Raspberry   1
```

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    cout << R"(Code Flavor
1 Chocolate
2 Vanilla
3 Strawberry
4 Raspberry
5 Butterscotch
Stock
1 20
2 15
3 15
4 20
5 20
)";

    int a, b;

    vector<int> v = {0, 20, 15, 15, 20, 20};
    vector<int> buy(6);
    vector<string> s = {"",
                        "Chocolate",
                        "Vanilla",
                        "Strawberry",
                        "Raspberry",
                        "Butterscotch"};

    while (cin >> a) {
        if (a == 0) {
            break;
        }
        cin >> b;

        // cout << "a,b " << a << " " << b << "\n";

        if (v[a] >= b) {
            v[a] -= b;
```

1.cpp C++17 / VS Code Dark+

```
            buy[a] += b;
        }

        cout << "Stock\n";
        for (int i = 1; i <= 5; i++) {
            cout << i << " " << v[i] << "\n";
        }
    }

    cout << "Cart\n";
    for (int i = 1; i <= 5; i++) {
        if (buy[i]) {
            cout << s[i] << " " << buy[i] << "\n";
        }
    }
}
```

CHAPTER 05 / PROGRAMMING PROJECT 14 · 19 / 24

電影評分推薦

Ch5 • Programming Project 14 AC 100 / 100

01 原題圖片 • Original Problem

14. You have collected reviews from four movie reviewers where the reviewers are numbered 0–3. Each reviewer has rated six movies where the movies are numbered 100–105. The ratings range from 1 (terrible) to 5 (excellent).

The reviews are shown in the following table:

	100	101	102	103	104	105
0	3	1	5	2	1	5
1	4	2	1	4	2	4
2	3	1	2	4	4	1
3	5	1	4	2	4	2

Write a program that stores this data using a 2D array. Based on this information your program should allow the user to enter ratings for any three movies. The program should then find the reviewer whose ratings most closely match the ratings input by the user. It should then predict the user's interest in the other movies by outputting the ratings by the reviewer for the movies that were not rated by the user. Use the Cartesian distance as the metric to determine how close the reviewer's

課本原題 • 第 270 頁 • 第 1/2 張

movie ratings are to the ratings input by the user. This technique is a simple version of the **nearest neighbor** classification algorithm.

For example, if the user inputs a rating of 5 for movie 102, 2 for movie 104, and 5 for movie 105, then the closest match is reviewer 0 with a distance of `sqrt` $((5-5)^2+(2-1)^2+(5-5)^2)=1$. The program would then predict a rating of 3 for movie 100, a rating of 1 for movie 101, and a rating of 2 for movie 103.

課本原題 • 第 271 頁 • 第 2/2 張

02 中文題目翻譯

你收集了四位影評者的評論，影評者編號為 0～3。每位影評者評了六部電影，電影編號為 100～105。評分從 1（很糟）到 5（非常好）。

評分如下表：

影評者對六部電影的評分

影評者	100	101	102	103	104	105
0	3	1	5	2	1	5
1	4	2	1	4	2	4
2	3	1	2	4	4	1
3	5	1	4	2	4	2

撰寫程式，用二維陣列儲存這些資料。根據這些資訊，程式應讓使用者輸入任意三部電影的評分。接著找出評分最接近使用者的影評者，並輸出該影評者對使用者尚未評分之電影的評分，用來預測使用者對其他電影的興趣。使用 Cartesian distance（笛卡兒距離）衡量影評者的評分與使用者輸入評分的接近程度。這項技術是 nearest neighbor classification algorithm（最近鄰分類演算法）的簡單版本。

例如，使用者給電影 102 評分 5、電影 104 評分 2、電影 105 評分 5，最接近的就是影評者 0，距離為：

> $\text{sqrt}((5-5)^2+(2-1)^2+(5-5)^2)=1$

程式接著預測電影 100 的評分為 3、電影 101 的評分為 1、電影 103 的評分為 2。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<vector<int>> v(200, vector<int>(200));
    string s =
        "315215\
421424\
312441\
514242";

    int it = 0;
    for (int i = 0; i <= 3; i++) {
        for (int j = 100; j <= 105; j++) {
            v[i][j] = s[it] - '0';
            it++;
        }
    }

    for (int i = 0; i <= 3; i++) {
    }

    // sqrt(pow(x,2)+pow(x,2)+pow(x,2));

    vector<int> c(200);
    int a, b;
    for (int i = 0; i < 3; i++) {
        cin >> a >> b;
        c[a] = b;
    }
```

1.cpp C++17 / VS Code Dark+

```cpp

    int mi = INT_MAX;
    int mi_i = 0;
    for (int i = 0; i <= 3; i++) {
        int sum = 0;
        for (int j = 100; j <= 105; j++) {
            if (c[j] != 0) {
                sum += pow(c[j] - v[i][j], 2);
            }
        }
        double sq = sqrt(sum);

        if (sum < mi) {
            mi = sum;
            mi_i = i;
        }
    }

    for (int j = 100; j <= 105; j++) {
        if (c[j] == 0) {
            cout << j << " " << v[mi_i][j] << "\n";
        }
    }
}
```

CHAPTER 05 / PROGRAMMING PROJECT 17 · 20 / 24

以陣列統計班佛定律

Ch5 • Programming Project 17 AC 100 / 100

01 原題圖片 • Original Problem

17. Programming Project 2.12 asked you to explore Benford's Law. An easier way to write the program is to use an array to store the digit counts. That is, `count[0]` might store the number of times 0 is the first digit (if that is possible in your data set), `count[1]` might store the number of times 1 is the first digit, and so forth. Redo Programming Project 2.12 using arrays.

課本原題 • 第 271 頁

02 中文題目翻譯

程式設計專題 2.12 要求你探索 Benford’s Law（班佛定律）。較簡單的寫法是用陣列儲存各首位數字的出現次數。也就是說，count[0] 可以儲存資料集中以 1 為首位數字的次數，count[1] 可以儲存以 2 為首位數字的次數，依此類推。使用陣列重做程式設計專題 2.12。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    ifstream fin;
    fin.open("numbers.txt");

    vector<double> v(10);

    string s;
    int total = 0;
    while (fin >> s) {
        v[s[0] - '0']++;
        total ++;
    }

    for (int i = 0; i < 10; i++) {
        if (i == 0) continue;
        cout << fixed << setprecision(8) << i << " " << v[i]/total*100 << "\n";
    }

    fin.close();
}
```

CHAPTER 06 / PROGRAMMING PROJECT 01 · 21 / 24

員工評等

Ch6 • Programming Project 1 AC 100 / 100

01 原題圖片 • Original Problem

1. Write a rating program for a software development firm with the following rating policies.
   a. Core competency – This parameter measures an employee's involvement, innovation, and passion toward their assigned tasks, and is graded on 10 points.
   b. Performance evaluation – This parameter measures an employee's performance over the year, and is graded on 30 points.
   c. Ideation – An employee is measured on their patent submissions, patents awarded, technical papers presented, etc. and is graded on 10 points.
   d. The performance evaluation accounts for 50% of an employee's final rating, core competency accounts for 30%, and ideation accounts for 20%.

   The total points obtained by an employee are summed up and normalized. Employees with points of 80 or more are rated a 1, those between 60 to 79 are rated a 2, those between 50 to 59 are rated a 3, and those less than 50 are rated a 4.

   The program will read an employee's points against the three criteria and output the total points obtained on a scale of 100, and the final rating point. Define and use a structure for the employee records.

課本原題 • 第 306 頁

02 中文題目翻譯

為一家軟體開發公司撰寫評等程式，採用以下政策：

a. 核心能力：衡量員工的投入、創新，以及對所指派工作的熱情等，以 10 分評分。

b. 績效評估：衡量員工整年的表現，以 30 分評分。

c. 創意發想：依提交的專利、獲得的專利、發表的技術論文等衡量，以 10 分評分。

d. 績效評估占員工最終評等的 50%，核心能力占 30%，創意發想占 20%。

將員工得到的分數加總並正規化。80 分或以上評等為 1；60～79 分評等為 2；50～59 分評等為 3；低於 50 分評等為 4。

程式讀入員工在三項標準得到的分數，並輸出滿分　100　分的總分與最終評等。請定義並使用結構來儲存員工紀錄。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp — C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

struct st {
    double a, b, c;
```

1.cpp C++17 / VS Code Dark+

```cpp
};

int main() {
    st m;
    cin >> m.a >> m.b >> m.c;
    double re = (m.a/10 * 30 / 100 + m.b/30 * 50 / 100 + m.c/10 * 20 / 100)*100.0;
    cout <<fixed<<setprecision(8) <<re;
    cout <<" "<<
    ((re>=80)?1:
    (re>=60)?2:
    (re>=50)?3:
            4);
}
```

CHAPTER 06 / PROGRAMMING PROJECT 07 · 22 / 24

Pizza 類別

Ch6 • Programming Project 7 AC 100 / 100

01 原題圖片 • Original Problem

7. Define a class called `Pizza` that has member variables to track the type of pizza (either deep dish, hand tossed, or pan) along with the size (either small, medium, or large) and the number of pepperoni or cheese toppings. You can use constants to represent the type and size. Include mutator and accessor functions for your class. Create a void function, `outputDescription( )`, that outputs a textual description of the pizza object. Also include a function, `computePrice( )`, that computes the cost of the pizza and returns it as a double according to the following rules:

   Small pizza = $10 + $2 per topping

   Medium pizza = $14 + $2 per topping

   Large pizza = $17 + $2 per topping

   Write a suitable test program that creates and outputs a description and price of various pizza objects.

課本原題 • 第 308 頁

02 中文題目翻譯

定義名為 Pizza 的類別，以成員變數記錄披薩種類（deep dish、hand tossed 或 pan）、尺寸（small、medium 或 large），以及 pepperoni 或 cheese 配料的份數。可以用常數表示種類與尺寸。為類別加入修改與存取函式。建立 void 函式 outputDescription()，輸出披薩物件的文字描述。另外加入 computePrice() 函式，依以下規則計算披薩價格並以 double 傳回：

> Small pizza = $10 + $2 per topping Medium pizza = $14 + $2 per topping Large pizza = $17 + $2 per topping

撰寫適當的測試程式，建立各種披薩物件，並輸出描述與價格。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

class Pizza {
   private:
    int type, size, pepperoni, cheese;
    double price;
    vector<int> v = {0, 10, 14, 17};
    vector<string> ty = {"", "deepdish", "handtossed", "pan"};
    vector<string> si = {"", "small", "medium", "large"};

   public:
    Pizza(int t, int s, int p, int c) : type(t),
```

1.cpp C++17 / VS Code Dark+

```
                                        size(s),
                                        pepperoni(p),
                                        cheese(c) {}

    void computePrice() {
        price = v[size] + 2 * pepperoni + 2 * cheese;
    }
    void outputDescription() {
        cout << fixed << setprecision(2);
        cout << ty[type] << " "
             << si[size] << " "
             << pepperoni << " "
             << cheese << " "
             << price << "\n";
    }
};

int main() {
    int a, b, c, d;
    while (cin >> a >> b >> c >> d) {
        Pizza pi(a, b, c, d);
        pi.computePrice();
        pi.outputDescription();
    }
}
```

CHAPTER 06 / PROGRAMMING PROJECT 10 · 23 / 24

Weight 類別

Ch6 • Programming Project 10 AC 100 / 100

01 原題圖片 • Original Problem

10. Create a class `Weight` that stores a weight internally in pounds. Create functions `setWeightPounds`, `setWeightKilograms`, and `setWeightOunces` that take an input weight in the specified weight scale, convert the weight to pounds if required, and store that weight in the class's member variable. Also, create functions that return the stored weight in pounds, kilograms, and ounces. Write a `main` function to test your class. Use the equations shown next to convert between the weight scales.

    1 pound = 16 ounces

    1 kilogram = 2.21 pounds

課本原題 • 第 308 頁

02 中文題目翻譯

建立 Weight 類別，內部以磅儲存重量。建立 setWeightPounds、setWeightKilograms 與 setWeightOunces 函式，接收指定單位的重量，必要時轉成磅，並儲存在類別成員變數中。另建立函式，分別以磅、公斤與盎司傳回所儲存的重量。為此類別撰寫 main 函式。使用下列等式換算不同重量單位：

> 1 pound = 16 ounces 1 kilogram = 2.21 pounds

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

class Weight {
   private:
    double p, kg, ou;

   public:
    void setWeightPounds(double in) {
        p = in;
        kg = in / 2.21;
        ou = in * 16;
    }
    void setWeightKilograms(double in) {
        in *= 2.21;

        p = in;
        kg = in / 2.21;
        ou = in * 16;
    }
    void setWeightOunces(double in) {
        in /= 16;
```

1.cpp C++17 / VS Code Dark+

```
23
24          p = in;
25          kg = in / 2.21;
26          ou = in * 16;
27      }
28      void pr() {
29          cout << fixed << setprecision(8);
30          cout << p << " " << kg << " " << ou << "\n";
31      }
32  };
33
34  int main() {
35      string a;
36      double in;
37      while (cin >> a >> in) {
38          Weight w;
39          if (a == "P") w.setWeightPounds(in);
40          if (a == "K") w.setWeightKilograms(in);
41          if (a == "O") w.setWeightOunces(in);
42          w.pr();
43      }
44  }
```

CHAPTER 06 / PROGRAMMING PROJECT 12 · 24 / 24

蔬果箱類別

Ch6 • Programming Project 12 AC 100 / 100

01 原題圖片 • Original Problem

12. Your Community Supported Agriculture (CSA) farm delivers a box of fresh fruits and vegetables to your house once a week. For this Programming Project, define the class `BoxOfProduce` that contains exactly three bundles of fruits or vegetables. You can represent the fruits or vegetables as an array of type `string`. Add accessor and mutator functions to get or set the fruits or vegetables stored in the array. Also write an `output` function that displays the complete contents of the box on the console.

    Next, write a `main` function that creates a `BoxOfProduce` with three items randomly selected from this list:

    ```
    Broccoli
    Tomato
    Kiwi
    Kale
    Tomatillo
    ```

    This list should be stored in a text file that is read in by your program. For now you can assume that the list contains exactly five types of fruits or vegetables.

    Do not worry if your program randomly selects duplicate produce for the three items. Next, the main function should display the contents of the box and allow the user to substitute any one of the five possible fruits or vegetables for any of the fruits or vegetables selected for the box. After the user is done with substitutions output the final contents of the box to be delivered.

課本原題 • 第 309 頁

02 中文題目翻譯

你的 Community Supported Agriculture（CSA，社區支持型農業）農場每週送一箱新鮮蔬果到家。這個程式設計專題中，請定義 BoxOfProduce 類別，內含恰好三束蔬果。可以使用 string 型別陣列表示蔬果。加入存取與修改函式，以取得或設定陣列中的蔬果。另外撰寫輸出函式，將箱內完整內容顯示在主控台。

接著，撰寫 main 函式，建立 BoxOfProduce，從以下清單隨機選出三項：

```
Broccoli
Tomato
Kiwi
Kale
Tomatillo
```

此清單應存放在程式會讀取的文字檔中。目前可假設清單恰好含五種蔬果。

若程式隨機選出的三項蔬果有重複，不用擔心。接著，main　應顯示箱內內容，並讓使用者以五種可能的蔬果之一，替換箱內任何已選出的蔬果。使用者完成替換後，輸出即將配送的蔬果箱最終內容。

03 我的 C++ 程式碼

Latest submitted source • syntax highlighted

1.cpp C++17 / VS Code Dark+

```cpp
#include <bits/stdc++.h>
using namespace std;

class BoxOfProduce {
   private:
   public:
};

int main() {
    srand(time(0));
    vector<string>item={"Broccoli",
        "Tomato","Kiwi","Kale","Tomatillo"};

    vector<string>v={item[rand()%5],
        item[rand()%5],
        item[rand()%5]};

    ifstream fin;
    fin.open("produce.txt");
    string s;

    vector<string>bag;
    while (getline(fin, s)) {
        bag.push_back(s);
    }

    cout<<"Initial\n";
    for(auto i:v){
        cout<<i<<"\n";
    }

    int a,b;
    while(cin>>a>>b){
        v[a-1] = bag[b-1];
    }


    cout<<"Final\n";
    for(auto i:v){
        cout<<i<<"\n";
    }



    fin.close();
}
```